SE(3)-WAM: A Design Study of History, Future Representation, and World–Action Interaction for Dynamic Manipulation Supplementary Material

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Research draft — results pending.
Red fields identify unverified settings, measurements, or study decisions.

This supplement expands the common protocol and the complete Q1–Q3 study: history encoding, future representation and prediction granularity, and world–action backbone organization and information flow. The selected SE(3)-WAM instance remains **[TBD]**. Mathematical descriptions of target $\mathrm{SE}(3)$, dense flow, and a one-way interface specify candidate implementations, rather than components already selected by the study. No result is inferred from an empty cell or from a historical experiment budget.

Table 1. Reading map. Main-paper section numbers refer to the separate main PDF.

Appendix	Contents	Main
A	Data, tasks, clocks, information, scoring	Sec. 3
B	Candidate implementation and mathematics	Secs. 3–4
C	Q1 history encodings and diagnostics	Sec. 5
D	Q2 representations and granularity	Secs. 4–5
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F	Cost, scale, and external protocols	Sec. 5
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A. Data and Dynamic-Manipulation Protocol

A.1. Task Families and Outcome Events

The controlled benchmark is intended to expose motion ambiguity, orientation, occlusion, contact timing, and adaptation after an external event. The task names below are inherited from the historical DB0 specification. The release, demonstration source, task subset, and evaluator configuration used for the formal Q1–Q3 comparison are **[MISSING: validated benchmark release and manifest]**. The implementation uses MuJoCo and robosuite as simulation infrastructure [5, 8]; their presence does not by itself establish the validity of a task or demonstration set.

Table 2. Working task-family definitions. These describe intended diagnostic contrasts; completed learning/evaluation results and the admitted release remain pending.

Family	Task and diagnostic distinction
P0	Forecast physical motion without robot execution; no action target.
T1-Belt	Intercept an object under continued conveyor motion.
T1-XYZ/SO3	Match centroid motion while changing whether asymmetric target orientation matters.
T1-OCC	Add physical occlusion to the moving-target task.
T2-TRANS/SE3	Release into a receptacle undergoing translation or rigid motion.
T3-Phase/Full	Traverse a rotating wire under phase-only or full task geometry.
T4	Capture a rolling sphere, sliding body, or roll–slip can leaving a ramp.
T5-Replan	Respond to one of two striker impacts and grasp the resulting keyed puck.

Success must follow the task's ordered physical events. A touch-only contact, pinning an object against a fixture, stopping the motion driver, or attaching an object before a grasp must not receive the same credit as a completed dynamic manipulation. The historical prehensile contract required opposing finger-pad contact for at least 50 ms, loss of support, 5 cm center-of-mass clearance, and a 0.5 s stable hold with the driver still moving. These numerical values are *historical candidate thresholds*; the validated evaluator thresholds and any task-specific exceptions are **[TBD]**. T2 additionally requires entrance crossing, collision-geometry containment, release, sufficiently low relative motion, and dwell; T3 uses calibrated wire-contact failure. Their exact geometric tolerances, dwell durations, and calibration

procedure are **[MISSING: evaluator configuration and validation evidence]**.

The event trace records the first failed requirement and its time. It supports separate examination of approach, first contact, grasp, transport or release, and post-impact adaptation. A failure label describes an observed event; assigning it to a forecasting or representation error requires an additional aligned prediction and action comparison.

A.2. Physical Time and Action-Dependent Labels

Let τ_t^s be the latest sensor capture time at a policy query, ℓ_t the capture-to-application delay, and Δ_a the high-level action interval. A world prediction evaluated at action index h refers to

$$\tau_h^\star = \tau_t^s + \ell_t + h\Delta_a. \quad (1)$$

History samples retain their actual timestamps and validity, so an equal frame count need not imply an equal time window. The physical horizon, history span, and sampling interval are reported separately for every comparison. The historical design used 20 Hz observations/high-level control and 500 Hz physics. These rates are not asserted as the final study configuration; accepted rates and delay support are **[TBD]**.

For a pose label between stored physics states, translation can be interpolated in the world or robot-base frame and rotation along the group geodesic:

$$\boldsymbol{R}(\alpha) = \boldsymbol{R}_j \operatorname{Exp}\big(\alpha \operatorname{Log}(\boldsymbol{R}_j^\top \boldsymbol{R}_{j+1})\big), \quad 0 \le \alpha \le 1. \quad (2)$$

The selected interpolation frame, sample rate, and label-validity tolerance must be part of the data configuration. Symmetry acts in the loss or metric, rather than changing a stored physical trajectory.

A robot can alter the future it is asked to predict. An unconditional target is valid only over a window in which unmatched robot contact has not changed the scene. After such contact, a label is masked unless the prediction is conditioned on the same stored high-level actions, controller, and delay that generated the label. P0 samples have no action target and contribute zero action loss. For each world-target family, the retained pre-contact windows, action-conditioning status, and fraction of invalidated labels are **[MISSING: target-validity audit]**. This distinction applies equally to pose, flow, point, RGB, and latent futures.

A.3. Policy Inputs and Privileged Information

The visual track uses sensor observations and robot state. The historical DB0 packet contained synchronized fixed and wrist RGB-D views, robot/gripper state, and capture/calibration metadata. History, history-valid masks, and past-action memory are constructed by a method-side loader. The final camera resolutions, calibration convention, proprioceptive fields, and allowed action memory are **[TBD]**. A method is not entitled to a field merely because it is stored in the same demonstration file.

Table 3. Information audit to accompany each study configuration. "Pending" requires checking the actual model input rather than its name.

Field	Visual policy	GT/oracle
RGB-D and calibration	Declared sensor input	Input use declared
Robot and gripper state	Declared current and past fields	Matched common state
Object pose and motion	Labels or evaluation only	Explicit privileged input
Instance masks	No undeclared mask input	Explicit oracle if used
Future states	Training labels only	Separate future oracle
Action memory	Loader choice: **[TBD]**	Matched and reported **[TBD]**
Task or goal input	**[TBD]**	

Object and goal poses, instance masks, contact state, future state, motion parameters, material, random-generator state, and event tapes are privileged records used by teachers, supervision, or evaluation. A ground-truth-state comparison is explicitly labeled an information/oracle track. It cannot be reported as visual localization or visual forecasting. In particular, GT kinematics can test representation or history questions under known state, but does not remove the need for a separate visual comparison.

Normalization, feature calibration, and uncertainty calibration use the training partition. A visibility mask derived from simulator identity is a label; it cannot silently become a deployable attention mask. Early-history padding is marked invalid instead of being represented as repeated evidence.

A.4. Pairing, Splits, and Replay

An exact-current group matches current scene appearance, object pose, robot state, and task parameters while retaining histories with different motion. The pairing audit must verify the matched variables and quantify any residual differences. History reversal or swapping is interpretable only when the substituted history belongs to the declared information boundary and current state remains fixed. The accepted construction and group counts are **[TBD]**.

Source episodes and all descendants of a paired/counterfactual group remain in one split. The split unit, train/validation/test counts, object families, motion support, spatial support, and duplicate filtering are **[MISSING: split manifest summary]**. Out-of-distribution tests vary predeclared object, spatial, or motion/physics factors and are excluded from checkpoint selection. Unannounced post-query impacts test adaptation; they are not charged as deterministic forecast error before the event

is observable.

The historical replay record stores two-view RGB-D, robot state, capture/frame alignment, and one high-level action per step; episode metadata records rates, controller, fixed delay, calibration, and versions. It does not export separate issued/applied action copies or a dense torque trace as policy inputs. A replay test must establish the actual recorded state, event order, and trajectory tolerance for the release used here. Its accepted tolerance and pass rate are **[TBD]**. Asynchronous capture, availability, inference, issue, and application logs belong to the separate systems profile in Appendix F.

A.5. Metrics and Statistical Units

Task success rate (SR) is reported with safety outcomes and task-specific event completion. The historical draft also used hierarchical task completion (TC) and success-conditional completion time (CT); their final aggregation, normalization, and units are **[TBD]**. Completion time is not averaged over successful and failed trajectories as if failures had completed. Contact-time error, relative contact-pose error, and collision measures link prediction to the physical action being evaluated; their definitions and threshold policies are **[TBD]**.

The proposed paired difference between methods m and b on matched cells is

$$\Delta_{m,b} = \frac{1}{N}\sum_{i=1}^{N}\left(y_{m,i} - y_{b,i}\right), \tag{3}$$

where y is the declared episode outcome and the cell/task weighting is fixed before test evaluation. Repeated trajectories from one source group are not independent draws. The historical plan used Wilson intervals for individual SRs and a paired, source-group-clustered bootstrap for differences, with seed-level values also reported. Its five headline seeds and 100 paired rollouts per cell are planned counts, not completed evidence. The realized seed count, rollout count, resampling procedure, confidence level, and weighting are **[MISSING: accepted statistical configuration]**.

B. Candidate Mathematics and Implementation

B.1. Shared Interface and Coordinate Convention

Each study candidate maps a declared history $\mathcal{H}_t$ to an action chunk, optionally through a future representation $\mathcal{Y}_t$. Comparisons record the history timestamps, input fields, prediction times, action times, parameter count, and inference budget. Q1 changes history encoding, Q2 changes $\mathcal{Y}_t$, and Q3 changes backbone organization and world–action paths. The phase-specific reference configuration and retained alternatives are **[TBD]**; no SE(3), flow, or one-way module is mandatory across the space.

For the geometric candidate below, $\boldsymbol{T}_O^W$ maps object-frame coordinates into a declared reference frame W. W is the robot base when using the historical depth-lifted implementation. Camera extrinsics map measured depth into W at the actual capture time. $\boldsymbol{T}_E^W$ is the end-effector transform. Translations use meters internally; tables may report centimeters. Rotation units and tangent-vector ordering are stated wherever values are reported. Changing reference frame requires changing both input coordinates and labels, rather than rotating an image feature while leaving supervision fixed.

For depth d at calibrated image location $\tilde{\boldsymbol{u}}$, the candidate lift is

$$\boldsymbol{x}^W = \boldsymbol{T}_C^W \begin{bmatrix} d\boldsymbol{K}^{-1}\tilde{\boldsymbol{u}} \\ 1 \end{bmatrix}, \tag{4}$$

with homogeneous normalization understood. The implementation must specify how patch depth is selected, how invalid depth is masked, and how moving-camera extrinsics are synchronized. These details are **[TBD]**.

B.2. Historical Metric-Encoder Candidate

The previous manuscript used a frozen DINOv3 encoder [4] with a 14×14 patch grid, depth-lifted patch positions, view identifiers, true time offsets, frame age, and validity bits. A factorized transformer performed within-frame spatial fusion followed by temporal fusion at matched patch/view locations. Robot/gripper state and past actions were separate tokens. These choices define one candidate, not the selected Q1 encoding. The actual backbone version, token counts, layer widths, attention order, trainable parameters, and handling of missing depth are **[TBD]**.

The historical MIN tier was four layers at width 256 with 224^2 images. Its optimizer was AdamW with learning rate 10^{-4}, cosine decay, 500 warm-up steps, and effective batch 64. These are retained for traceability as *unverified candidate settings*; Table 4 separates them from the final configuration. A new comparison cannot inherit a checkpoint's training exposure and describe it as a matched fresh initialization.

B.3. SE(3), Flow, and Symmetry Candidates

A target-pose decoder can predict a latest pose $\hat{\boldsymbol{T}}_t$ and a sequence of tangent displacements $\hat{\boldsymbol{\xi}}_h = [\hat{\boldsymbol{\rho}}_h^\top, \hat{\boldsymbol{\phi}}_h^\top]^\top$:

$$\hat{\boldsymbol{T}}_h = \mathrm{Exp}(\hat{\boldsymbol{\xi}}_h^\wedge)\hat{\boldsymbol{T}}_t. \tag{5}$$

This is a left-multiplied, reference-frame increment. A body-frame increment would instead multiply on the right and must be implemented consistently. The rotation metric for declared object symmetry group $\mathcal{S}$ is

$$e_R = \min_{\boldsymbol{S}\in\mathcal{S}} \left\| \mathrm{Log}(\hat{\boldsymbol{R}}_h^\top \boldsymbol{R}_h \boldsymbol{S})^\vee \right\|_2. \tag{6}$$

The symmetry set is task- and object-specific. A visually symmetric body with an asymmetric grasp feature must not be assigned a symmetry that erases the task's orientation

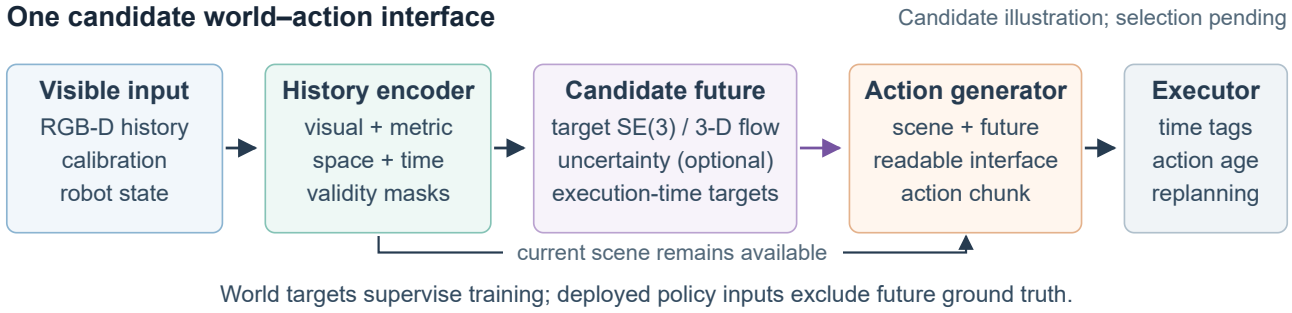


Figure 1. The inherited physical-interface candidate, shown as a conceptual implementation option. Target $\mathrm{SE}(3)$, context flow, and a one-way action interface are replaceable choices in Q2 and Q3; this diagram does not identify the model selected by the study.

Table 4. Implementation configuration record. The center column preserves the old candidate; the final column must be filled from the actual study.

Item	Historical candidate	Formal study
Image input	Two RGB-D views, 224^2	[TBD]
Encoder	Frozen DINOv3	[TBD]
Temporal module	Four layers, width 256	[TBD]
Observation/physics	$20/500$ Hz	[TBD]
World horizon	Eight targets, 50–400 ms	[TBD]
Action chunk	Eight E7 steps	[TBD]
Optimization	AdamW, 10^{-4}, batch 64	[TBD]
Initialization	Staged candidate training	[TBD]
Data exposure	Not established in manuscript	[TBD]
Hardware/precision	Not established in manuscript	[TBD]
Checkpoint rule	Validation selection required	[TBD]

requirement. The object list and symmetry definitions are [TBD].

A metric-flow decoder represents displacement of reference-frame points, $\boldsymbol{f}_{i,h} = \boldsymbol{x}_i(\tau_h^\star) - \boldsymbol{x}_i(\tau_t^s)$. Historical simulated labels transported visible depth samples by persistent body identity. Undefined correspondences and action-dependent futures are masked according to Appendix A. The target-only rigid displacement implied by pose is

$$\hat{\boldsymbol{f}}_{i,h}^{\mathrm{rigid}} = \left(\hat{\boldsymbol{T}}_h \hat{\boldsymbol{T}}_t^{-1}\right) \boldsymbol{x}_i - \boldsymbol{x}_i, \tag{7}$$

where homogeneous coordinates are implicit. Consistency with this quantity can be measured or trained only on valid target points, with the region definition supplied as supervision or a declared predicted mask. It cannot use an undeclared privileged target mask at inference.

Pose and flow expose different candidate trade-offs: rigid state makes target orientation explicit, while distributed motion can represent non-target constraints. The Q2 study determines when these properties affect actions; their mathematical availability is not evidence of a task benefit. Point, trajectory, RGB, and latent alternatives use their own reconstruction or prediction metrics alongside the same action-level outcomes.

B.4. Losses and Target Normalization

The old uncertainty candidate used diagonal log variance $\boldsymbol{s}$ and residual $\boldsymbol{e}$ with

$$\mathcal{L}_{\mathrm{het}}(\boldsymbol{e}, \boldsymbol{s}) = \tfrac{1}{2} \sum_j \left[\exp(-s_j) e_j^2 + s_j\right]. \tag{8}$$

Residual units, translation/rotation scales, log-variance bounds, and any calibration transform are [TBD]. Calibration is assessed on held-out training/validation data and must not be fitted on the final test outcomes. An uncertainty head is optional; a scalar prediction error is not automatically a calibrated uncertainty estimate.

For the pose–flow candidate, a possible objective is

$$\mathcal{L} = \mathcal{L}_a + \lambda_T \mathcal{L}_T + \lambda_F \mathcal{L}_F + \lambda_C \mathcal{L}_C. \tag{9}$$

Each loss is normalized by its number of valid targets. P0 has $\mathcal{L}_a = 0$; unavailable history, occluded correspondences, and invalid post-contact futures do not become zero-valued observations. The actual weights, normalization, target masks, and effective contributions per task are [MISSING: candidate-specific loss configuration]. These terms do not define the objective of every Q2 representation or Q3 backbone.

The action family can use regression or conditional flow matching [2]. With source chunk $\boldsymbol{a}_0$, demonstrated chunk $\boldsymbol{a}_1$, and conditioning $\boldsymbol{c}_t$, the latter trains

$$\begin{aligned} \boldsymbol{a}_\tau &= (1-\tau)\boldsymbol{a}_0 + \tau \boldsymbol{a}_1, \\ \mathcal{L}_a &= \|\boldsymbol{v}_\theta(\boldsymbol{a}_\tau, \tau \mid \boldsymbol{c}_t) - (\boldsymbol{a}_1 - \boldsymbol{a}_0)\|_2^2. \end{aligned} \tag{10}$$

Source distribution, action normalization, integration schedule, number of function evaluations, and sampling seed convention are [TBD]. The shared action objective is selected independently of claiming a future-representation benefit and is held fixed within the corresponding contrast.

B.5. Candidate Bridge and Training Stages

The historical one-way candidate reduced future outputs to horizon-indexed tokens. One possible summary combines end-effector-relative predicted pose, velocity, pooled flow, uncertainty, and physical target time:

$$\boldsymbol{z}_h = g_\psi\big(\operatorname{Log}((\boldsymbol{T}_E^W)^{-1}\hat{\boldsymbol{T}}_h), \hat{\boldsymbol{v}}_h, \operatorname{Pool}_i\, w_{i,h}\hat{\boldsymbol{f}}_{i,h}, \hat{\boldsymbol{\sigma}}_h, \tau_h^\star\big). \tag{11}$$

Every supplied quantity must be computable from the declared policy inputs or predictions. In particular, a ratio involving time to feasible contact cannot silently use teacher feasibility at deployment. Token dimensions, pooling, velocity construction, and feasibility information are [TBD]. The Q3 study compares this route with other backbones and interactions.

The old staged recipe first trained world prediction, then selected an action-only objective, then compared auxiliary world supervision with detached or end-to-end consumption. It is a candidate schedule; the final stage durations, trainable modules, initialization source, and checkpoint rule are [TBD]. Shared RGB-D augmentation must transform calibration and geometric labels consistently; color changes must not alter metric depth. Cached features, if used, must correspond to the actual encoder, crop, and calibration of the comparison. Hardware, total optimizer steps, effective examples, and cache use are reported per variant so that training reuse does not become unreported extra data or compute.

C. Q1: History Encodings and Diagnostics

C.1. Reference Conditions and Encoding Comparison

Q1 asks how finite history should be encoded to support motion understanding, future prediction, and action. A history encoding is evaluated under a declared future target and world–action reference, rather than under an assumed final SE(3)/flow pipeline. The phase-specific reference, encoder candidates retained after screening, input fields, history span, and sampling intervals are [MISSING: Q1 comparison configuration]. The formal result table must identify whether the compared encoders see the same samples and differ only in organization, or whether an explicit sampling/window axis also changes.

A current-image control is distinguished from a strict memory-free control. If the policy retains end-effector history, past actions, a recurrent state, or previous predicted chunks, it still has memory even when visual history is removed. The historical label H0V therefore denotes a visual-history intervention only; its retained robot/action fields must be listed. GT object history is an information diagnostic and appears in a separate track from visual-history encodings.

Table 5. Q1 configuration record. Placeholder rows reserve separate comparisons; they do not assert that a candidate has been implemented or run.

Comparison	Required common conditions	Unresolved configuration
Encoding organization	Same input samples, future target, action objective	[TBD]
Window/sampling	Same visible fields, declared physical span and gaps	[TBD]
Visual-history removal	List retained EEF/action memory	[TBD]
Strict memory-free	No hidden recurrence or cached history	[TBD]
GT-history diagnostic	Matched GT track; never pooled with visual SR	[TBD]

C.2. Motion Ambiguity, Occlusion, and Sampling

Exact-current comparisons examine whether histories distinguish motion that is ambiguous in a current frame. A direction diagnostic matches present pose and appearance while changing the preceding motion. The construction must be audited before learning results are interpreted: residual pose, visibility, or robot-state cues could otherwise explain a difference. Accepted pair counts and current-state matching tolerances are [TBD].

Occlusion comparisons report the observed history interval, occlusion duration, and visibility at the query and predicted execution time. Removing older frames and changing temporal spacing are separate manipulations. The study reports seconds as well as frame count, and keeps the action horizon fixed where the question concerns only history. For irregular histories, actual time offsets and masks are retained instead of relabeling samples at uniform intervals.

History swapping tests the effect of substituted temporal evidence with current inputs held fixed. A useful outcome must connect changes in predicted motion, the selected action, task success, and safety in the direction implied by the paired scene. Probe predictability or action sensitivity alone cannot show that historical information caused a useful action improvement. The history-necessary subset is validated independently of the aggregate SR; its acceptance evidence

Table 6. Pending Q1 results. SR differences use the corresponding matched reference within a track; GT and visual inputs remain separate.

Track/diagnostic	SR (%)	Δ SR	Unsafe (%)
Visual encoding comparison	[TBD]	[TBD ± CI]	[TBD]
Strict memory-free comparison	[TBD]	[TBD ± CI]	[TBD]
Direction-ambiguous subset	[TBD]	[TBD ± CI]	[TBD]
Occluded subset	[TBD]	[TBD ± CI]	[TBD]
Sampling-gap comparison	[TBD]	[TBD ± CI]	[TBD]
GT-history diagnostic	[TBD]	[TBD ± CI]	[TBD]

is [TBD].

C.3. Expanded Results and Reporting

Table 6 reserves the expanded summary. The final table is indexed by actual encoder, task, and seed rather than by invented candidate names. For each comparison, the per-task/seed source, number of matched episodes, interval calculation, and failure exclusions are [TBD].

Prediction errors are reported at the physical times used for action, with translation and orientation distinguished when those quantities exist. Action outcomes additionally include contact-time error, relative pose at contact, safety, and task completion. Their available result arrays are [MISSING: Q1 per-task, per-seed and per-horizon results]. No positive effect of longer history or a particular encoder is established by the present placeholders.

The selected Q1 encoding is provisional until Q2 determines which future information is useful. After changing the future representation, retained history candidates are revisited under the new reference; the completed cross-checks and any ranking changes are reported in Appendix E.

D. Q2: Future Representation and Granularity

D.1. Representation Scope and Comparison Axes

Q2 compares what the future representation describes and at what granularity. The reference history encoding and information flow are provisional selections from the preceding phases: [MISSING: Q2 reference configuration]. All representations use declared physical prediction times and the same available input information within a contrast. Changes in resolution, target count, prediction objective, and online computation are recorded, rather than being hidden in a representation name.

The working space includes target-only, context-only, uniform-granularity, and heterogeneous-granularity futures. "Target" means the manipulated object or task-relevant local state; "context" includes non-target movers, obstacles, support/receptacle geometry, and free-space constraints relevant to feasible action. The exact region construction and deployable target assignment are [TBD]. A privileged region mask is supervision or an oracle condition, not an unreported visual input.

Table 7. Q2 comparison specification. The actual admitted representation, resolution, and measured costs remain pending for each row.

Axis	Controlled question	Required declaration
Target-only	Is target future sufficient for this task?	Target state/region; omitted context
Context-only	Can surrounding motion/constraints guide action?	Context state; current target access
Uniform	What follows from one common prediction granularity?	Space/time resolution and token budget
Heterogeneous	Does allocating detail differently help?	Target/context allocation and total cost
Representation	Which future state exposes useful action information?	SE(3), flow, points, trajectory, RGB/latent as admitted

"Target fine, relevant context coarse" is a falsifiable hypothesis. It is not encoded as the selected architecture before the comparison. If it lacks a stable benefit, the study reports the tasks, horizons, and budgets where the effect is absent or negative, and chooses the final instance accordingly. The completed candidate set and reasons for excluding any candidate are [MISSING: Q2 candidate admission record].

D.2. Horizon, Region, and Budget Matching

Prediction horizon is measured from capture time to the time the action takes effect, as in Eq. (1). Spatial density and temporal density are independent axes: more points at one instant are not more future time samples. For every candidate, the reporting record includes the physical time grid, region support, number of predicted elements, feature dimensions, trainable parameters, prediction NFE if applicable, and end-to-end latency.

Where exact parameter and latency equality cannot both hold, the comparison states which budget is matched and supplies the measured alternative cost. An advantage caused by denser temporal samples, more parameters, a longer history, or lower action age cannot be assigned to the semantic representation alone. Resolution/cost curves, if measured, report the actual points; missing points are not connected into an invented curve. The accepted budget envelope and completed resolution settings are [TBD].

The geometric candidate of Appendix B uses rigid pose for the target and metric displacement for distributed motion. Point and trajectory candidates must specify their correspondence and sampling rules; RGB and latent candidates must specify decoding and the action-visible interface. A com-

Table 8. Pending Q2 action-level expansion. Each final row is a measured candidate under the stated reference; this table currently records no ranking.

Future scope	SR (%)	Contact error	Unsafe (%)
Target-only	[TBD ± CI]	[TBD]	[TBD]
Context-only	[TBD ± CI]	[TBD]	[TBD]
Uniform granularity	[TBD ± CI]	[TBD]	[TBD]
Heterogeneous granularity	[TBD ± CI]	[TBD]	[TBD]
Strong history baseline	[TBD ± CI]	[TBD]	[TBD]
Declared future oracle	[TBD ± CI]	[TBD]	[TBD]

mon action interface can be used when that is the controlled question, but its bottleneck capacity and training exposure must be reported. Changing both the representation and the backbone is reserved for an explicit interaction check rather than attributed to Q2 alone.

D.3. Prediction Measures and Action-Level Measures

Representation-specific errors describe whether the chosen target is predicted well. For a rigid target, translation error and symmetry-aware rotation error are reported separately. Flow uses valid-point EPE in metric units and can include target rigidity consistency. Point, RGB, and latent errors are not numerically pooled with pose error. The available per-horizon values are **[X.XX]** cm, **[X.XX]**°, **[X.XX]** cm, and **[X.XX]** for the historical geometric fields; the full arrays and the applicable candidate mapping remain **[TBD]**.

The action-level comparison uses the same task outcome definitions for all representations. Contact timing and relative contact pose connect prediction to interception and grasping. Feasible approach, receptacle containment, and collision/safety outcomes test context-dependent constraints. No scalar prediction metric substitutes for task success. A lower average error could still miss the few times or regions that determine contact.

The contact-error field in Table 8 is expanded into separate timing, translation, and orientation columns when actual results are available; the units and aggregation must then be explicit. Task/seed/horizon breakdowns, confidence intervals, and safety denominators are **[MISSING: Q2 complete result arrays]**. A main-paper claim about context requires evidence on the relevant constraint, not only a lower global flow error.

D.4. Oracles, Calibration, and Failure Strata

Future oracles estimate headroom under explicitly privileged information. A future-XYZ oracle and a future-SE(3) oracle are different controls; both remain distinct from a vision model. The historical gap-closed diagnostic is

$$G_{\text{oracle}} = \frac{\text{SR}(P) - \text{SR}(B)}{\text{SR}(O) - \text{SR}(B)}, \tag{12}$$

for predictive method P, matched history baseline B, and declared oracle O. It is interpretable only when the denominator supports a meaningful comparison; the raw differences and intervals are reported regardless. The pending historical field is **[XX]**%; oracle definition, coverage, and the measured denominator remain **[TBD]**.

Calibration is reported only for candidates with a probabilistic output and a defined calibration procedure. Error and uncertainty can be stratified by visibility, speed, acceleration, orientation change, horizon, and event phase when those labels are available. A comparison before and after robot contact also states whether it is action-conditioned. Failures that reverse a representation ranking stay in the main argument, with the full breakdown here: **[MISSING: Q2 negative results and conditional effects]**.

After Q3 changes the interaction backbone, the leading and contrasting Q2 candidates are rechecked. Representation selection and backbone selection are therefore provisional until the measured interactions are known, rather than three independent choices assumed to compose.

E. Q3: Backbones, Information Flow, and Cross-Checks

E.1. Backbone Study and Component Diagnostics

Q3 studies how world and action computation is organized and how information passes between them. A component ablation within one one-way model does not replace this backbone study. Each admitted configuration records token organization, shared/separate parameters, attention directions, training losses, action-visible futures, and gradient paths. The completed backbone families, matched parameter envelope, and reference Q1/Q2 configuration are **[MISSING: Q3 backbone comparison configuration]**.

The historical manuscript supplied three local mechanism conditions. They remain useful candidate diagnostics under shared current context $\boldsymbol{X}$, world target, and action objective:

$$\begin{aligned} &\text{Auxiliary: } \boldsymbol{a} \sim p_\theta(\boldsymbol{a} \mid \boldsymbol{X}), \quad \boldsymbol{z} \sim q_\psi(\boldsymbol{z} \mid \boldsymbol{X}), \\ &\text{Detached: } \boldsymbol{a} \sim p_\theta(\boldsymbol{a} \mid \boldsymbol{X}, \text{sg}[\boldsymbol{z}]), \\ &\text{End-to-end: } \boldsymbol{a} \sim p_\theta(\boldsymbol{a} \mid \boldsymbol{X}, \boldsymbol{z}). \end{aligned} \tag{13}$$

Here sg blocks action gradients through the future interface; it does not imply removal of the world loss. The auxiliary condition has training-time world supervision but no online action edge from $\boldsymbol{z}$. These three conditions do not exhaust jointly generated or differently organized world–action models. Their architectural scope must remain explicit.

Table 9. Q3 reporting fields. Auxiliary, online, and joint-generation properties are recorded separately for each actual backbone.

Field	Required record
Backbone	Token organization, shared layers, separate experts
Information	Allowed attention and conditioning at train/test time
Supervision	World/action losses and valid-label masks
Gradient	Which action gradients reach which world/shared modules
Generation	Parallel, sequential, or coupled world/action computation
Budget	Parameters, training exposure, NFE, measured action age
Initialization	Fresh run or declared checkpoint continuation

E.2. Interventions and Their Interpretation

A direct intervention is defined only when a model exposes a meaningful future interface. Holding current perception, robot state, and action-source noise fixed, the historical interface allows replacement by zero, a matched-sample swap, calibrated noise, or an oracle future. For a different backbone, the intervention point and the input fields it changes must be specified anew; an inapplicable intervention is marked inapplicable, rather than simulated by changing arbitrary internal activations.

A swap must preserve the declared current-state match and target-time alignment. Noise has a stated distribution, scale, and calibration set. Zeroing may be out of distribution and is interpreted with the matched training/removal control where available. An oracle supplies privileged future information and does not establish deployable visual accuracy. These interventions assess whether the online path affects actions; useful causal use additionally requires task outcomes, safety, and a coherent direction of change. Action sensitivity by itself is insufficient.

The inherited placeholder fields are swap action change **[X.XXX]**, swap SR change **[-X.X]** percentage points, and zero/noise SR changes **[-X.X]**/**[-X.X]** percentage points. Their intervention definitions, sample counts, and confidence intervals remain **[TBD]**. They cannot yet support a claim that the one-way bridge or explicit future is causally useful.

E.3. Expanded Backbone Results

Per-task and per-seed outcomes, world-quality changes under action gradients, optimization behavior, module cost, and parameter counts are **[MISSING: Q3 complete results]**. The final report distinguishes an improvement due to representation shaping, online prediction use, coupled generation, or lower execution age only where the corresponding controlled comparison supports that distinction.

Table 10. Pending Q3 expansion. Final rows name actual backbone/flow configurations; auxiliary-only and online paths are not conflated.

Configuration/contrast	SR (%)	Δ SR	Age (ms)
Backbone organization study	**[TBD]**	**[TBD ± CI]**	**[TBD]**
Auxiliary supervision only	**[TBD]**	**[TBD ± CI]**	**[TBD]**
Detached online future	**[TBD]**	**[TBD ± CI]**	**[TBD]**
End-to-end online future	**[TBD]**	**[TBD ± CI]**	**[TBD]**
Admitted joint-generation path	**[TBD]**	**[TBD ± CI]**	**[TBD]**
Applicable intervention	**[TBD]**	**[TBD ± CI]**	**[TBD]**

Table 11. Required cross-check record. Only completed combinations enter the evidence; the table does not authorize or imply a full factorial experiment.

Revisit	Question	Result/configuration
Q1 after Q2	Does the new future target change history needs?	**[TBD]**
Q2 after Q3	Does representation ordering change with information flow?	**[TBD]**
Selected instance	Does the composed model retain the measured gains?	**[TBD]**
Negative interaction	Which combination reverses or removes a gain?	**[TBD]**

E.4. Cross-Checks and Selecting the Final Instance

The progression is a common protocol and baseline with initial Q3 screening, then Q1, Q2, deeper Q3 comparison, and a limited set of cross-checks before model selection. A phase winner is conditional on its reference; it is not automatically the best component after the other axes change.

The selected SE(3)-WAM configuration is **[TBD]**. Its final input boundary, history, future representation, backbone, action objective, and deployment cost are filled only after these comparisons. A null or negative effect on one axis narrows the design conclusion; it does not invalidate reporting the complete study. Results that materially change the ordering are summarized in the main paper rather than appearing only in this supplement.

F. Cost, Scale, and External Evaluation

F.1. Fixed Delay and Measured Systems Latency

A registered fixed-delay condition in simulation is distinct from the wall-clock latency of an implementation. The former tests behavior under a known observation-to-action offset; the latter measures the actual sensing, preprocessing,

Table 12. Pending systems profile. Numbers are not inferred from nominal simulation rates or fixed-delay settings.

Quantity	p50	p95
Sensor/preprocessing time (ms)	[TBD]	[TBD]
History/world computation (ms)	[TBD]	[TBD]
Action generation time (ms)	[TBD]	[TBD]
End-to-end loop time (ms)	[XX.X]	[XX.X]
Applied action age (ms)	[TBD]	[XX.X]
Peak device memory	[TBD]	
Deadline-miss rate (%)	[X.X]	

computation, queueing, and command-application path. An offline trajectory does not establish p50/p95 runtime or real-system action age.

For a command applied at τ^{apply} from an observation captured at τ^{s}, action age is $\tau^{\text{apply}} - \tau^{s}$. Runtime profiling records capture, availability, inference start/end, issue, and application timestamps where those events can be measured. Hardware, precision, batch size, warm-up, number of timed samples, synchronization, and camera/control rates are **[MISSING: reproducible profiling configuration]**. The profile states which stages are included in each reported latency.

The historical continuous-execution candidate tags an action chunk by source observation and target control steps. A newest-wins executor discards elapsed actions and replaces overlapping future steps with a completed newer chunk. This is a candidate deployment mechanism; the actual executor, queue policy, deadline, and dropped-command behavior are **[TBD]**. Comparing predictors under this executor requires reporting realized action age as well as model-forward time.

F.2. Scale and Training Cost

Model scale and data scale are separate studies. A parameter-count comparison records trainable and frozen components, while a data-size comparison records source episodes and effective training exposure. Both retain the relevant split and evaluation protocol. Actual model sizes, data subsets, training steps, accelerator-hours, memory, and replicated seeds are **[MISSING: completed scale configurations and measurements]**. An unrun scale setting is not a point on a scaling curve.

Candidate decoders with different NFE or online future computation are compared on both quality and realized cost. A Pareto claim requires measured alternatives under stated budgets; lower parameter count alone does not show lower latency. Any benefit that disappears after matching action age or training exposure is reported with that condition.

Table 13. External protocol record. Historical subset names and trial counts are proposals; no transfer result is established by this table.

Track	Configuration to establish	Result
DOMINO single arm	Native subset, interface, split, adaptation	[TBD]
DOM-Dyn4	Native DynamicVLA comparison and scorer	[TBD]
PhysMani	Included normal/high families and scorer	[TBD]
LIBERO static	Task subset and static adaptation protocol	[TBD]
Real conveyor	Calibration, conditions, interventions, trial allocation	[TBD]

F.3. Native External Protocols

External evaluation is kept separate from the controlled benchmark. The historical manuscript proposed a single-arm DOMINO subset [1], a DOM-Dyn4 comparison with DynamicVLA [6], PhysMani families [7], static LIBERO compatibility [3], and a real conveyor task. These are candidate tracks. The accepted versions, task lists, adaptation rules, native baselines, and completed runs are **[TBD]**.

Each track reports its native scores and safety definitions. Scores are not pooled into controlled-benchmark SR. Baselines need the corresponding native code and scoring, with any retraining or adaptation stated. A single-arm adaptation is not described as the unchanged full benchmark. Static compatibility does not by itself validate dynamic forecasting, and an external dynamic score does not identify which Q1–Q3 mechanism helped.

F.4. Real-System Evidence

The old budget of 180 Franka conveyor trials is a historical plan, not a completed count. The actual trial allocation, object set, speed/delay ranges, order randomization, calibration, action interface, failure criteria, and human interventions are **[MISSING: accepted real-system protocol]**. Results must state the realized number of independent trials, exclusions, safety terminations, and uncertainty. Physical success, pose-estimation quality, and end-to-end runtime are measured separately.

No transfer benefit, real-conveyor success, or static non-regression is established by the current placeholders. Any external evidence supporting an abstract-level claim also receives a result summary in the main paper; this appendix

holds protocol detail and disaggregated outcomes.

G. Failures, Aligned Visualizations, and Video

G.1. Prediction and Execution at the Same Physical Time

A qualitative comparison uses the same physical episode and query time for all candidates. Rows identify methods or interventions; columns identify capture time and the physical prediction/application times. Predicted and actual states are aligned to those times, rather than to equally spaced video frame indices. Camera, reference frame, units, object identity, and visibility are stated in the caption. A pose overlay uses measured/predicted geometry and the actual calibration, not an illustrative reconstruction presented as a measurement.

The draft currently contains no accepted rollout cases. Source episodes, prediction arrays, application timestamps, and figure exports are **[MISSING: real aligned qualitative assets]**. Empty statistical or trajectory panels are not replaced by synthetic curves or imagined successes. A conceptual diagram, when useful, is labeled as a schematic and carries no measured performance implication.

G.2. Case Selection and Failure Attribution

Case selection is declared before a visual claim is made. The final set should cover the main supported comparisons and failures that qualify their scope, using a reproducible episode-selection rule or an explicitly illustrative selection. The selection rule, eligible episode pool, and selected identifiers are **[TBD]**. A visually clear success cannot establish a general success rate, and a chosen failure cannot establish its population frequency.

Event traces first identify what happened: approach missed a feasible contact, contact occurred at the wrong time or orientation, a grasp was not maintained, a release failed containment, a collision occurred, or adaptation followed an external impact. Whether the cause was perception, prediction, feasibility, action generation, or stale execution is tested against synchronized outputs and the appropriate comparison. These categories are diagnostic questions, not a measured taxonomy. Their frequencies and attribution evidence are **[MISSING: annotated failure results]**.

Representation-specific failures remain visible. Examples include a target prediction that omits a relevant obstacle, a context representation that lacks grasp orientation, or a prediction whose average error hides a contact mistake. These are hypotheses for inspection, not claimed observations. Actual counterexamples and negative results, if they alter the Q1–Q3 conclusion, are summarized in the main paper with the full aligned record here.

Table 14. Media/asset index to fill from actual exports. No videos are asserted as delivered by this draft.

Item	Required content	File/case reference
Q1 diagnostic	Matched current state and different histories	**[TBD]**
Q2 comparison	Predicted/actual future and contact outcome	**[TBD]**
Q3 intervention	Fixed current input; declared future/path change	**[TBD]**
Negative case	Full context and relevant event trace	**[TBD]**
External/real	Native task, time scale, and interventions	**[TBD]**

G.3. Supplementary Media Index

Submission media are provided as anonymous supplementary files. The index records filenames, durations, speed relative to real time, associated configuration, and the result/case they illustrate. Videos disclose cuts, slow motion, resets, and human intervention where present. A caption links a clip to a measured episode; the clip does not replace quantitative evidence or an essential method description in the main paper.

The final media audit checks author/institution identifiers, screen overlays, file metadata, audible identity cues, caption consistency, and whether the files open at the supplied path. The actual file inventory and completed audit are **[TBD]**. Supplementary references below identify the sources used in this document; they are independent of the main paper's bibliography.

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